

1. The base of a rectangle has a length of 8. One of the whole numbers greater than 5.2 and less than 7.3 is selected at random for the height of the rectangle. Find the probability that the area of this rectangle is greater than 50. Express your answer as a common fraction reduced to lowest terms.
2. If $f(x) = |3x - 12|$, then an equation of a line of symmetry is $x = k$. Find the value of k .
3. Give the smallest possible **positive** value of x for which $\cos(2x^\circ - 60^\circ) = 0$.
4. The cubic equation whose solution set is $\{1, 5, 11\}$ can be written as $x^3 + kx^2 + wx + p = 0$. Find the value of $(k + w + p)$.
5. The first term of a geometric sequence of real terms is 405. The fifth term of this sequence is 20480. Find the third term of this sequence.
6. A circle with center at $(-6, 3)$ contains the point $(2, -12)$. An arc of the circle has measure of $35^\circ 15'$. Find the area of the sector of the circle that is bounded by two radii and the arc. Express your answer as a **decimal** rounded to the nearest tenth.
7. One of the transformations necessary to produce the graph of $y = 2x^2 + 24x - 1$ from the graph of $y = x^2$ is a vertical shift k units downward. Find the value of $|k|$.
8. The function g satisfies the functional equation $g(x) + g(y) = g(x + y) + 2xy - 8$ for every pair (x, y) of real numbers. If $g(1) = 22$, find $g(-11)$.